

COMP9444 Project Summary

Neural Networks & Deep Learning: NLP Classification of FOMC Texts for Market Analysis

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I. Introduction

Central bank communication, notably Federal Open Market Committee (FOMC) statements have had a significant impact on global financial markets. Investors, policymakers and others closely observe these announcements to predict upcoming market movements and make informed decisions accordingly. There have been numerous academic literatures and models exploring the extraction of sentiment or stances (i.e. hawkish, dovish, or neutral), but struggle to capture the nuances and intricacies of financial policy statements.

The aim of this project is to explore multiple models from academic literature and industry, assess the relative complexities, and ultimately improve upon them to drive more accurate sentiment analysis. Our primary goal is to **develop a “best practice model” that supplants current models**, and in turn its outputs – sentiment scores – can be utilised for policy analysis, economic forecasting, or financial market modelling. For example, Correa et al. (2020) found that it was a useful predictive tool for banking crises, with a 1%-point increase in negative sentiment linked to 29% increase in the likelihood of crisis. Similarly, Hilscher et al. (2024) demonstrates that sentiment forecasts policy rates multiple Taylor Rule quarters in advance, while Deng et al. (2012) shows sentiment correlation with S&P Returns. As such, this sentiment analysis project has been found to be useful in many areas, including but not limited to predicting financial stability and monetary policy decisions.

Our contributions include the exploration, development and evaluation of multiple Natural Language Processing (NLP) models, using Long Short-Term Memory (LSTM) as a foundation, Bidirectional Encoder Representations from Transformers (BERT), Robustly Optimized BERT Pretraining Approach (RoBERTa), DeBERTa (Decoding-enhanced BERT with disentangled attention), and [QwenLM](#), an architecture developed by AliBaBa. We explore and assess these models within the context of their capabilities and limitations and build upon them by fine-tuning the models to handle the complexities inherent in central bank communication.

Furthermore, in the spirit of “developing a best practice” we seek to assess the merits of stacking models to overcome the shortcoming of individual models and provide a more robust and reliable sentiment classification tool that handles these complexities more effectively. As existing models are commonly observed as “black boxes” with limited explainability into its decision-making, we seek to incorporate Local Interpretable Model-Agnostic Explanations (LIME) into our model to ensure to foster greater confidence in the model’s predictions, and in turn, also make it more practical for real-world applications.

II. Related Work

Traditional financial sentiment analysis relied on dictionary-based approaches, such as Loughran and McDonald's (2011) work using customised financial word lists to “assign” values to certain words like “growth” as positive, but these approaches lacked the fundamental ability to handle context-dependent language that limited an understanding of financial language nuances, but found that 73.8% of words classified as negative by the Harvard Dictionary were not considered negative in financial contexts. Therefore, capturing context-dependent language was understood to be crucial for achieving accurate sentiment classification in financial domains, that was enabled with later models.

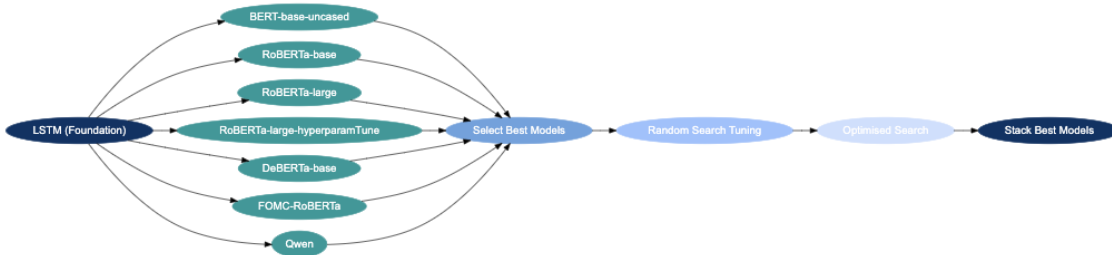
A breakthrough came with the introduction of transfer learning models LSTM networks that allowed models to capture context-dependent language over longer sequences and understand dependencies and nuances in contexts. This is crucial, as Araci (2019) did find that it performed exceptionally well (for the time), at an accuracy of 71% on the Financial PhraseBank dataset. The limitations in handling dependencies, was addressed later with Transformers. Later advancements in transformer-based architectures drove models like Bidirectional Encoder

Representation from Transformers (BERT), and significantly improved contextual language understanding. Notably, in the same study from Araci (2019), he did produce a higher accuracy with FinBERT at an 86% accuracy. While 86% is high, it still faced challenges with dependencies in more specialised domains like finance, where He et al. (2021) propose Decoding-enhanced BERT with Disentangled Attention (DeBERTa), which enhances the attention mechanism by enabling the content and position to take a step backwards, and incorporate absolute positional embedment, and therefore enable the model to comprehend the structural nuances and relationship of sentences and relationship between word combinations. In the context of finance, Yang et al. (2022) used DeBERTa to financial texts, and reported better accuracy – but suffered from issues like computational costs, and need for large datasets. Considering the computation costs, more compact yet effective architectures like Alibaba Groups’ Qwen-7B were also assessed, and is a 7-billion parameter transformer model, where Han (2023), using Quantised Low-rank Adaptation has shown promising results for sentiment analysis in FOMC datasets, with an evaluated accuracy of 82.27%.

More recent literature has explored the potential of model stacking in the context of sentiment analysis, and Etelis et al. (2023), demonstrated a mean accuracy of >90% across 10 datasets, through stacking via a HEC algorithm. Notably it does demonstrate the potential effectiveness of such stacking strategies that we seek to explore. But notably it is extremely complex and computationally expensive. While significant process has been made with transformer-based models in sentiment analysis, challenges remain in balancing accuracy, efficiency and the ability to generalise to nuances. The methodology, through exploring all these strategies, seek to add to existing literature, with the hopes of deployment in real-world applications.

III. Methods (0.5)

Our methodology focuses on evaluating and enhancing existing NLP models to build a “best practice model” that successfully classifies sentiment, and capturing the nuances of Central Bank communications, with a diverse set of models from LSTM, BERT, DeBERTa, RoBERTa variants, FOMC-RoBERTa, Qwen and other variants, based on proven effectiveness showcased in prior research, where sought to fine-tune the models, and evaluate the potential for stacking. Except for LSTM, all are pre-trained models, chosen for proven performance from the rigorous academic research conducted, where seek to surpass existing literature and attempts by others.



The dataset was compiled and cleaned already, and our pre-processing only involved tokenisation, standardisation, removal of irrelevant characters, on top of lemmatisation. A comparison of multiple architectures was evaluated using the Precision, F-1 score, and the accuracy for the narrowing of models. Although attempts to rebalance were investigated like SMOTE, and adding class weight balance in the training process, it did not yield ideal results. This is likely that SMOTE introduces artificial noise leading to degraded performance, and the class weight adjustments, did lead to overfitting, with severely reduced performance across three metrics. We did conduct some research and concluded that it would be unrealistic to rebalance.

Our process of hyperparameter tuning mimics the process demonstrated by Bergstra and Bengio (2012) with limitations of computationally expensive models, by having a two-pronged process of random search, and then more focused search. For example, the DeBERTa Random Search process for 10 combinations took 25 hours alone on an RTX3080. As such, random search was initially used to identify promising regions, focused by more focused searches like grid search or Bayesian optimisation (Hutter, Kotthoff, & Vanschoren 2019).

Incorporating model stacking to leverage the strengths of models is a well-established technique, and academic literature has found enhanced predictive performance and robustness. This approach is beneficial in such complex tasks like sentiment analysis, where different models may capture various aspects of the data, where we sought to combine best models, where we developed Meta-models. We then attempt to address the black-box nature of deep learning models to enhance transparency by using Local Interpretable Model-Agnostic Explanations, in explaining individual predictions by approximating the model locally – and in turn, which word or phrases contributes most to prediction, for example comparing predictions from LSTM and RoBERTa.

IV. Experimental Setup (0.5)

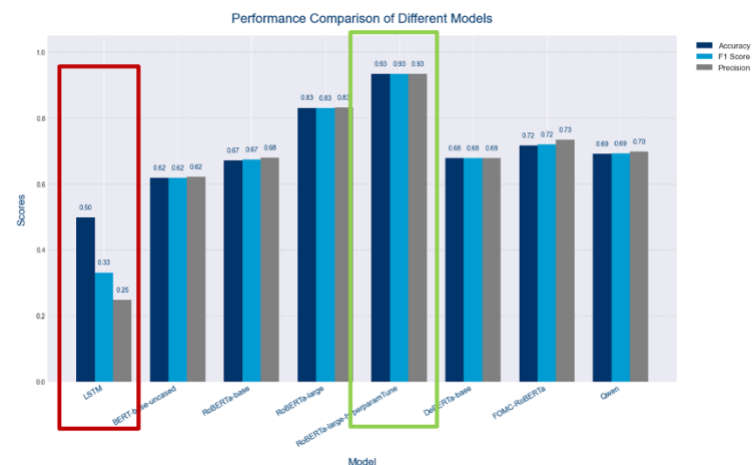
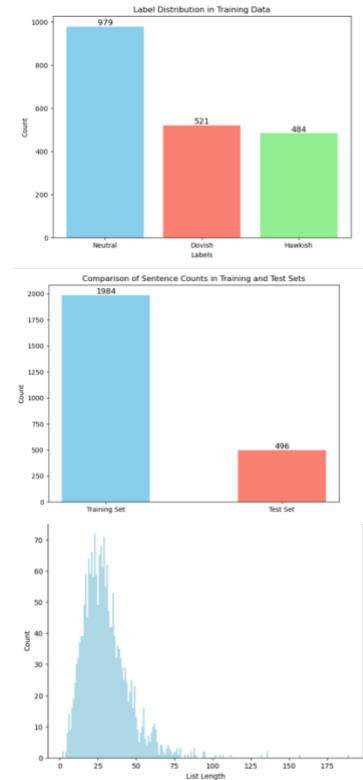
The dataset provided was from the Financial Services Innovation Lab of Georgia Tech on [GitHub](#), with multiple training and testing datasets, where we chose “5768” due to its balance between computational feasibility and relative balances across the three classes ({0, 1, 2}), with a 2:1:1 split. Attempts to rebalance faced issues, with SMOTE introducing more noise.

Notably in the data exploration, we noticed that most lists have fewer than 50 tokens, with a peak at about 20-30 tokens, which is critical for in the data pre-processing stage. For feature representation where we used Word2Vec for LSTM and Tokenisation for BERT models. Word2Vec converts words into continuous vector representations and understand sequences of varying lengths, whereas BERT uses sub-word tokenisation methods to handle rare words to break it down into smaller units.

We used an 80/20 split for training and testing, our models were evaluated using Accuracy, F1 Score and Precision metrics that enables an understanding of overall classification performance, while accounting for class imbalances. In the process of hyperparameter tuning, we mostly utilised a randomised search at the outset, focusing on key hyperparameters like Learning Rate, Batch Size, Epochs, Max Length, Hidden and Attention Dropout. Additionally, the algorithm has a patience limit of 2 epochs for early stopping, if not improvement in validation was observed, ultimately helping optimise training efficiency, while preventing overfitting. Following of which the best 2-3 models from the evaluation exercise, were further fine-tuned with optimised or grid search, and then a stacking process to create meta-models was implemented. The same evaluation metrics were used to examine performance.

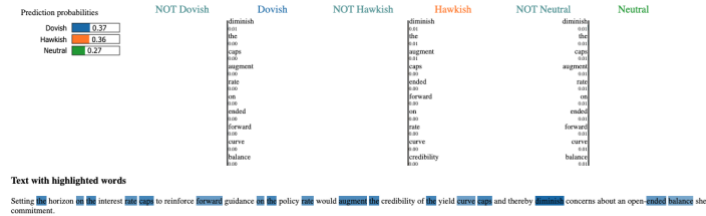
V. Results (0.5)

As detailed below, the performance of individual fine-tune transformer models performed in relative terms, far greater than the benchmark we had set with LSTM. Our 93% metric across Accuracy, F1 and precision for the RoBERTa-Large, was an excellent result. Because of limited public information, the only industry example we know of, is Wellington Management’s Quant Investment Group that utilised ChatGPT4o API and generated a ~97% accuracy based on personal conversations. So, our attempt certainly outperformed Araci’s FinBERT, and closed the gap between industry and our models. On the other hand, as seen, the attempts at stacking result did not yield desirable results, at less than <50% across all three-evaluation metrics.

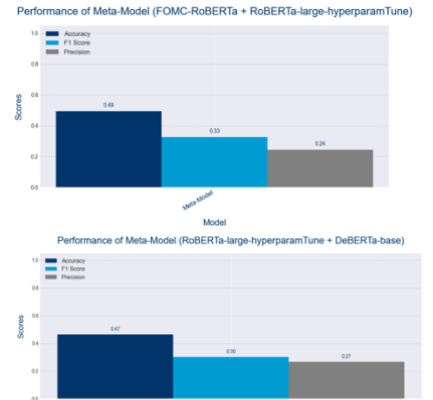
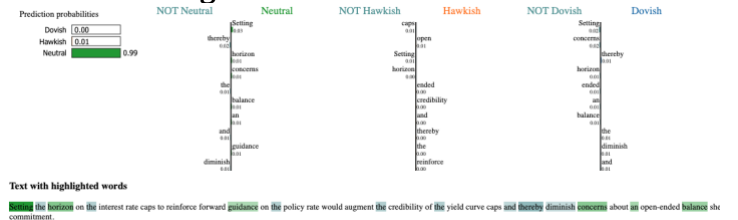


The stacking process, while initially promising based on the studies we evaluated. It appears that the issue was our model selection that had similar architectures in terms of transformer-based attention mechanisms, in the case of RoBERTa-large, DeBERTa and FOMC-RoBERTa. This lack of diversity might have limited the potential benefit by combining complementary strengths, and amplifying exist errors from the base models. On the brighter side, our LIME diagram added some explainability, with a comparison of LSTM and RoBERTa in the below case showcasing more uncertainty in distributions for the LSTM predictions, in contrast to RoBERTa-large’s certainty in “Neutral” Sentiment.

LSTM LIME:



RoBERTa-large LIME:



In terms of real-world deployment, our RoBERTa-large-hyperparamTune model, with a consistent 93% performance, shows potential, but a gap does exist with the use of Wellington’s GPT-40 API at 97% accuracy. It does show room for improvement. Current literature like Araci’s FinBERT, that showed lower performance at around ~ 85% accuracy, our approach demonstrates how newer transformer models and tuning, can significantly boost performance for sentiment classification. Unlike Araci’s work, our evaluation incorporates attempts to combine transformers into a stacked model, notwithstanding issues, it does highlight an area for future exploration.

VI. Conclusions

Our development of a sentiment analysis model for FOMC text with RoBERTa-Large at 93% accuracy, F1 Score and precision narrows the gap between academic models and industry standards. This does demonstrate the innate potential of current state-of-the-art transformer architectures in performing sentiment analysis, and drastically outperforming more traditional models like LSTM.

The proposed solution leveraged a two-staged hyperparameter tuning process (random, then more tailored search), to yield much better performance than what the literature had suggested, at 93%. A comprehensive experimentation process of stacking demonstrated that for Sentiment Analysis, at least in the way in which we conducted, was not successful. However, our fine-tuned model did outperform older models like FinBert and does illustrate the impact of newer architectures. Our implementation of LIME also added explainability to our results, which is a crucial strength over other model. As with limitations, we suspect the poor performance of the meta-models is due to the similarity amongst the base models in architectures. Further, high computational costs made hyperparameter tuning exceptionally challenging that drastically limited the breadth of our exploration.

On the other hand, if given more time and resources, we would likely pivot our focus on to transformer models that provide distinct contextual understanding mechanisms, like Bloom, or perhaps GPT-3. These would bring more diversity our predominantly BERT-centred models and can enhance the diversity of the ensemble. In consideration of the considerable computational requirements for large transformer models, we have considered techniques like knowledge distillation, that helps transfer knowledge from larger pre-trained models to a smaller model while maintaining similar levels of accuracy. Studies like Hinton et al. (2015) have demonstrated the feasibility. Overall, all contributions do show promise for real-world sentiment analysis in financial texts and does almost close the gap to industry performance standards, and despite not reaching Wellington’s Quantitative Investment Group’s ~ 97% benchmark, it does provide a foundational basis for future research and improvement.

VII. Bibliography

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